GG-S-770B
August 26, 1976
SUPERSEDING
Int. Fed. Spec. GG-S-00770A(DSA-DM)
October 25, 1968 and
Fed. Spec. GG-S-789
June 14, 1948

FEDERAL SPECIFICATION

STRIP, ABRASIVE, DENTAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic abrasive strips suitable for use during dental polishing procedures.

1.2 Classification.

1.2.1 Grit sizes. Dental abrasive strips covered by this specification shall be of one type, flint, and shall be of the following grit sizes as specified (see 6.1):

Grit sizes:

180
150
120

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6520

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Federal Specifications:

L-P-377 - Plastic Sheet and Strip, Polyester.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain extra copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by th Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kanasas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D882 - Testing for Tensile Properties of Thin Plastic Sheeting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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U.S. Department of Commerce Standard:

PS8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20234.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive. Abrasive shall be crushed from clean, high quality quartz or quartzite (flint) and shall be free from extraneous materials. There shall be three grit sizes of flint strips furnished: grit size 120, grit size 150, and grit size 180. The grit sizes shall conform to Product Standard PS8 for screen grit sizes (see 6.2).

3.1.2 Bonding agent. The bonding agent shall be a high quality, waterproof, resin type bonding agent suitable for the purpose intended.

3.1.3 Backing. The backing shall be polyethylene terephthalate (polyester) film without the addition of dyes or pigments complying with the requirements of type I, general purpose film of L-P-377, except that the haze requirements shall not apply. Dimensional requirement shall be in accordance with this specification. Tests for tensile strength and elongation shall be in accordance with 4.4.2.

3.2 Design and construction. The strips shall be flat, safe-sided, plastic-backed abrasive strips suitable for smoothing and polishing dental cement and plastic restorations in the oral cavity under both wet and dry conditions. The plastic backing shall be 0.0024 to 0.0035 inch thick per slit roll, and 6 inches ± 1/16 inch in length. One side of the strip shall be coated with abrasive grain in the grit size for the respective grades as specified in 1.2.1 and 3.1.1. The abrasive coating shall be of the closed coat type and bonded in such a manner as to uniformly and suitably cover the backing surface to produce a sharp cutting abrasive product. The abrasive shall wear off gradually when in use. The other side of the backing shall be uncoated. Strips shall be furnished in the grit size specified (see 6.1).

3.2.1 Width assortment. Unless otherwise specified, a box containing a minimum of 50 narrow strips, 3/32 + 1/32 inch -1/64 inch wide and a minimum of 50 medium strips, 5/32 + 1/32 inch -1/64 inch wide, shall constitute a unit of issue.

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3.3 Performance. The abrasive strips shall perform satisfactorily when both dry and wet (after a one-hour soak in distilled water). The strips shall be pliable yet shall possess sufficient stiffness, to be used during dental polishing procedures. The binder shall not fuze, glaze, or create heat when in use. The strips shall be pliable yet possess sufficient stiffness. The strips shall stay flat without warping under both normal operative procedures and after an oven heat treatment at 180 °F. for 50 minutes.

3.4 Workmanship. Workmanship shall be first class throughout. Abrasive strips shall be free from defects which either detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the abrasive, bonding agent, and backing material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II, III and IV (see 4.2.1, 4.2.2, 4.3, and 4.3.1) and 4.3.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one strip from each strip width.

TABLE I. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Acceptance number shall be zero for all sample sizes for grading, tensile strength and elongation, and performance tests. The acceptable quality level (AQL) for dimensional testing shall be 4.0 (percent defective).

TABLE II. Sampling criteria for tests

Component	Characteristic	Requirement	Test procedure	Unit of product	Insp. level
Raw abrasive	Grading	3.2	4.4.1	N/A	1/
Uncoated backing	Tensile strength and elongation	3.1.3	4.4.2	N/A	1/
Finished item	Performance	3.3	4.4.3	1 strip from each strip width	S-1

1/ Test may be performed either on raw material or on end item (by recovery procedures). When testing is done on raw material, sufficient amount from each lot of applicable component shall be taken for testing. When testing is done on end item, sufficient amount of applicable component shall be taken from each lot of end item by recovery procedures to perform applicable test.

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4.3 Examination. Strips shall be examined for defects, including, but not limited to, those listed in table III.

TABLE III. Classification of defects

Categories	Defects
Major A:	
101	Grit size not as specified.
102	Strip not coated with abrasive grain on one side.
103	Abrasive does not uniformly cover one surface of backing.
104	Extraneous materials such as dirt and lint present in abrasive.
105	Strips not flat, plastic-backed and safe-sided.
106	Cracks, chips, holes, cuts, or tears present in strips.
Major B:	
151	Count not as specified.
152	Thickness of plastic backing for strips not within specified tolerances.
153	Length of strips not as specified.
154	Size (width) not as specified.
Minor:	
201	Creases present in strips.
202	Foreign material present on safe side of strips.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

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TABLE IV. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.2 Examination of closure and reinforcement for containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix to that specification.

4.4. Tests.

4.4.1 Abrasive grading. The grading of the abrasive grit shall be tested in accordance with the test methods set forth in PS8.

4.4.2 Backing. The tensile strength and elongation of the backing (prior to coating) shall be tested in accordance with method A of ASTM D882. The specimen size shall be 1-inch wide by 5 inches long. Elongation rate shall be one hundred percent per minute. Separation between jaws shall be 2-inches. Specimen thickness shall be determined before testing.

4.4.3 Performance test. The performance of the strips shall be evaluated for properties specified in 3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Fifty narrow strips and fifty medium strips of one grit size only shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

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5.1.1.2 Intermediate package. Twenty-four unit packages containing strips of one grit size only shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages (288 units) containing strips of one grit size only shall be packed in an exterior container constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and banding shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages (288 units) containing strips of one grit size only shall be packed in an exterior container constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grit size required (see 1.2.1).
(c) If the unit of issue is different from that specified (see 3.2.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Grit sizes. Screen grit sizes are those which are coarse enough to be measured and controlled by the use of sieves.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6520-1412

Civil Agency Coordinating Activities:

VA-DMS
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